# A Scaling Invariant at Direction-Field Creases in 3D Navier–Stokes Reconnection: Conditional Regularity Failure, Emergent Balance, and Lean 4 Formalisation

Adam Roediger* Patrick Moriarty†

April 2026

**Abstract**

We study the structure of the viscous term $\nu\nabla^2\omega$ at direction-field creases in 3D incompressible Navier–Stokes vortex reconnection. Decomposing vorticity as $\omega = |\omega|\,\xi$, the product rule produces $|\omega|\nabla^2\xi$—a term whose regularity depends on the direction field $\xi = \omega/|\omega|$. At the reconciliation surface between antiparallel vortex tubes, $\xi$ develops a crease: a thin layer where the direction field rotates through 180° between coherent tubes.

**Pseudospectral DNS at** $\mathrm{Re} = 6400$ **(**$N = 64, 128, 256$**) shows that the Lipschitz constant** $\Lambda = \sup|\nabla\xi|_F$ **scales as** $N^{0.86\pm0.05}$ **(**$R^2 > 0.96$**), consistent with** $O(1/\Delta x)$**.** However, this growth occurs where $|\omega| \to 0$—the reconciliation gap between tubes. **The product** $\sup(|\omega| \cdot |\nabla\xi|_F)$ **is resolution-independent:** $\gamma = 0.04 \approx 0$ **across a** $4\times$ **resolution range (coefficient of variation** $< 3\%$ **at** $t = 0, 10$**).** The individual factors scale as $N^{+\beta}$ and $N^{-\beta}$ at the location of the product maximum, an interface ring between the tube cores and the crease centre.

We formalise a conditional implication in Lean 4 with Mathlib: if $\Lambda$ diverges *and* $|\omega| > 0$ at the crease, then $\nu\nabla^2\omega$ exits $L^2$. The formal proof (13 theorems, 0 `sorry`, 10 Z3-certified bounds) is mathematically valid. *The numerical data shows that the two antecedents are coupled through the product invariant rather than independently realised—whether they can be jointly instantiated is an open question requiring late-reconnection field data at higher resolution.*

The formal verification comprises 13 obligations, 10 Z3-certified numerical bounds, and 21 definitions grounded in experimental data via exact rational arithmetic. All source code, certificates, and DNS data are included as ancillary files.

**MSC 2020:** 35Q30, 68V15, 76D05 **Keywords:** Navier–Stokes, regularity, vorticity direction field, scaling invariant, vortex reconnection, formal verification, Lean 4

## 1 Introduction

The question of whether smooth solutions to the 3D incompressible Navier–Stokes equations remain smooth for all time, or whether finite-time singularities can form, remains one of the central open problems in mathematical fluid dynamics [9, 16].

Classical blowup criteria characterise singularity formation through scalar norms: the Beale–Kato–Majda criterion [1] requires $\int_0^T \|\omega\|_{L^\infty}\,dt = \infty$, while Constantin and Fefferman [2] showed that regularity is controlled by the vorticity direction field $\xi = \omega/|\omega|$—specifically, Lipschitz

*Live Oak Legacy Builders LLC.
†Bitumen Machina LLC, Coastal Georgia.

continuity of $\xi$ in regions of high vorticity suffices to prevent blowup. The Constantin–Fefferman–Majda geometric criterion [3] refined this to a local condition on the alignment of vorticity with the strain eigenvectors.

Recent progress on the closely related Euler equations has dramatically sharpened the picture. Building on Hou and Luo's numerical discovery of a potential singularity [14], Chen and Hou [4] established finite-time blowup for the 2D Boussinesq and 3D axisymmetric Euler equations via a nearly self-similar mechanism. Chen and Hou [5] subsequently gave a rigorous computer-assisted proof of finite-time singularity for 3D Euler with smooth initial data and boundary. Elgindi [10] independently established finite-time blowup for $C^{1,\alpha}$ Euler data (a lower regularity class). For Navier–Stokes, Tao [18] constructed finite-time blowup solutions for an averaged version of the equations, demonstrating that the known conservation laws do not by themselves prevent singularity. Hou [12] investigated nearly self-similar blowup candidates for generalised axisymmetric Navier–Stokes, and Chen, Hou, and Huang [6] recently gave a computer-assisted proof of nonuniqueness of Leray–Hopf solutions to the unforced 3D Navier–Stokes equations.

Our contribution starts from the Constantin–Fefferman direction-field framework but shifts the focus from regularity criteria to direct measurement of the direction field's structure under grid refinement. The product rule decomposition $\omega = |\omega|\xi$ exposes how $\xi$'s regularity enters the viscous term: $\nu\nabla^2\omega$ contains $|\omega|\nabla^2\xi$, which becomes distributional if $\xi$ loses smoothness while $|\omega|$ remains positive. We formalise this conditional implication and then ask: what does the direction field actually do at the crease?

**The answer is an emergent scaling balance.** At the reconciliation surface between antiparallel vortex tubes, $\xi$ must rotate through 180° over a thin layer. The Lipschitz constant $\Lambda = \sup|\nabla\xi|_F$ scales as $O(1/\Delta x)$—the crease sharpens with resolution. But the vorticity magnitude at the crease location simultaneously decays, so that the product $\sup(|\omega| \cdot |\nabla\xi|_F)$ remains resolution-independent. This is distinct from, but complementary to, the nonlinear blowup scenarios studied by Chen–Hou and Elgindi: those works identify how the nonlinear term drives singularity; we characterise the scaling structure of the viscous term at the crease.

**What we prove. Under three hypotheses—that $\Lambda$ diverges, that $|\omega| > 0$ at the crease, and that a Gagliardo–Nirenberg type estimate relates $\|\nabla^2\xi\|_{L^2}$ to $\Lambda$—the viscous term is unbounded in $L^2$, and the solution exits $H^2$ regularity.** The formal proof in Lean 4 covers the algebraic core (13 theorems, 0 `sorry`). This conditional theorem is mathematically valid regardless of the numerical evidence.

**What we observe.** Pseudospectral DNS of antiparallel vortex tube reconnection at Re = 6400 ($N = 64, 128, 256$) on $[0, 2\pi]^3$ shows:

(a) $\Lambda$ scales as $N^{0.86\pm0.05}$ (consistent with $O(1/\Delta x)$);

(b) the product $\sup(|\omega| \cdot |\nabla\xi|_F)$ scales as $N^{0.04} \approx N^0$ (resolution-independent);

(c) the crease is localised in the reconciliation gap between tubes, where $|\omega|$ is small;

(d) 100% of crease voxels have direction-field gradients sharper than the local Kolmogorov scale ($\Lambda_\eta > 1$).

*The conditional theorem's antecedents (H1: $\Lambda$ diverges; H2: $|\omega| > 0$ at the crease) are coupled through the product invariant rather than jointly realised.*

[OPEN: Whether the product invariant is a barrier to singularity (the flow self-regulates) or a plateau before transition to a different regime (the invariant breaks at later times or higher Re) is the central open question. See Section 6.]

**Formal verification.** The conditional implication is formalised in Lean 4 [15] with Mathlib [17], comprising 13 formally verified theorems with no `sorry` gaps. Numerical bounds are certified by the Z3 SMT solver [19] in the QF_LRA fragment. The Lean formalisation covers the algebraic structure; the analytic content (the Gagliardo–Nirenberg type estimate linking $\|\nabla\xi\|_{L^\infty}$ to $\|\nabla^2\xi\|_{L^2}$) is stated as an assumption. See Section 5 for trust boundaries.

# 2 Setup and Definitions

## 2.1 Vorticity direction field

Consider the 3D incompressible Navier–Stokes equations in vorticity form on the periodic domain $[0, 2\pi]^3$:

$$\partial_t \omega = (\omega \cdot \nabla)u + \nu \nabla^2 \omega, \tag{1}$$

where $\omega = \nabla \times u$ is the vorticity, $\nu > 0$ is kinematic viscosity, and $u$ is recovered from $\omega$ via the Biot–Savart law.

**Definition 2.1** (Direction field)**.** In regions where $|\omega| \neq 0$, define the vorticity direction field

$$\xi = \frac{\omega}{|\omega|}.$$

This is a unit vector field ($|\xi| = 1$) encoding the local axis of rotation.

**Definition 2.2** (Lipschitz constant)**.** The Lipschitz constant of $\xi$ at time $t$ is

$$\Lambda(t) = \sup_{x \in \Omega,\, |\omega(x)|>0} |\nabla\xi(x)|_F,$$

where $|\cdot|_F$ denotes the Frobenius norm. When $\Lambda(t) < \infty$, the direction field is Lipschitz continuous.

**Definition 2.3** (Criticality ratio)**.** Define the dimensionless criticality ratio

$$K(t) = \Lambda(t) \cdot \eta(t), \qquad \eta(t) = \left(\frac{\nu^3}{\varepsilon(t)}\right)^{1/4},$$

where $\eta$ is the Kolmogorov microscale and $\varepsilon$ the energy dissipation rate. $K > 1$ means the direction field varies on scales finer than the dissipation scale.

## 2.2 Product rule decomposition

Since $\omega = |\omega|\,\xi$, the Laplacian decomposes as:

**Proposition 2.4** (Viscous term decomposition)**.**

$$\nabla^2\omega = |\omega|\,\nabla^2\xi + 2\,(\nabla|\omega| \cdot \nabla)\xi + \xi\,\nabla^2|\omega|. \tag{2}$$

*Proof.* Direct application of the product rule $\nabla^2(fg) = f\nabla^2 g + 2\nabla f \cdot \nabla g + g\nabla^2 f$ with $f = |\omega|$ and $g = \xi$. □

The three terms have distinct regularity properties:

(i) $|\omega|\,\nabla^2\xi$: singular if $\xi$ develops a crease ($\nabla^2\xi$ becomes a distribution);

(ii) $2(\nabla|\omega| \cdot \nabla)\xi$: bounded if $|\omega| \in H^1$ and $\nabla\xi \in L^\infty$;

(iii) $\xi\,\nabla^2|\omega|$: bounded if $|\omega| \in H^2$ (since $|\xi| = 1$).

## 3 Main Theorem

**Hypothesis 1** (Direction crease formation)**.** The Lipschitz constant of the direction field diverges: $\forall M > 0$, $\exists\, t$ such that $\Lambda(t) > M$.

**Hypothesis 2** (Vorticity persistence at crease)**.** There exists $c > 0$ such that $\inf_{x \in \mathcal{C}(t)} |\omega(x,t)| \geq c$ for all $t$, where $\mathcal{C}(t)$ is the crease region (the set where $|\nabla \xi|$ is maximal).

**Hypothesis 3** (Analytic estimate)**.** There exists $c' > 0$ such that $\|\nabla^2 \xi\|_{L^2(\mathcal{C})} \geq c'\Lambda$ when $\Lambda = \|\nabla \xi\|_{L^\infty} \to \infty$.

Hypothesis 3 is a Gagliardo–Nirenberg type estimate relating the $L^\infty$ norm of $\nabla \xi$ to the $L^2$ norm of $\nabla^2 \xi$ on the crease support. **We state this as an assumption; it is not proved in this paper or in the Lean formalisation.** The estimate is plausible when the crease region has bounded geometry (the gradient concentrates on a surface of bounded area, forcing the second derivative to grow), but a rigorous derivation requires detailed knowledge of the crease structure.

**Theorem 3.1** (Viscosity regularity failure)**.** *Under Hypotheses 1, 2, and 3, the viscous term $\nu \nabla^2 \omega$ is unbounded in $L^2$:*

$$\neg\, \exists\, B > 0 \;\; \forall\, t : \quad \|\nu \nabla^2 \omega\|_{L^2} \leq B.$$

**Remark 3.2** (Terminology)**.** We use "viscosity bootstrap failure" to describe the mechanism by which the viscous term's regularity requirements are self-referential. This is not a bootstrap argument in the PDE continuation/extension sense (as in Tao's continuity method [18]), but rather a *regularity presupposition*: evaluating $\nu \nabla^2 \omega$ at the crease requires $\nabla^2 \xi \in L^2$, which is the regularity that viscous smoothing is supposed to maintain.

*Proof sketch.* From the product rule (2), the viscous term satisfies

$$\|\nu \nabla^2 \omega\|_{L^2} \geq \nu\, |\omega|_{\min} \cdot \|\nabla^2 \xi\|_{L^2(\mathcal{C})} - \nu\, B_{\text{cross}}, \tag{3}$$

where $|\omega|_{\min} \geq c > 0$ on the crease region $\mathcal{C}$ by Hypothesis 2, and $B_{\text{cross}}$ accounts for the bounded cross terms (ii) and (iii).

Applying Hypothesis 3, the bound (3) gives $\|\nu \nabla^2 \omega\|_{L^2} \geq c''\Lambda - B_{\text{cross}}$ for $c'' = \nu\, c\, c' > 0$. By Hypothesis 1, $\Lambda$ exceeds any bound, so $c''\Lambda - B_{\text{cross}}$ exceeds any bound. Formally: given any purported bound $B$, choose $t$ with $\Lambda(t) > (B + B_{\text{cross}} + 1)/c''$ (which exists by Hypothesis 1), yielding $\|\nu \nabla^2 \omega\|_{L^2} > B$. Contradiction. □

**Remark 3.3** (What the Lean proof covers)**.** The Lean formalisation (OBL-10, `viscosity_bootstrap_failure`) proves the final algebraic step: if a sequence $\Lambda(n) \to \infty$ and $\omega_{\min} > 0$, then $\omega_{\min} \cdot \Lambda(n) - B_{\text{cross}}$ is unbounded. This is essentially the Archimedean property of $\mathbb{Q}$. Note that the Lean proof works over $\mathbb{Q}$ (rationals), not $\mathbb{R}$—the Archimedean property holds for both, but the choice of $\mathbb{Q}$ means the formalisation operates in discrete arithmetic, not the continuous setting of the PDE. The PDE content—the product rule decomposition, the lower bound (3), and the analytic estimate (Hypothesis 3)—is stated in the Lean doc-strings but not formalised as Lean theorems. A full formalisation would require Lean libraries for Sobolev spaces and distributional derivatives that do not yet exist in Mathlib.

**Remark 3.4** (The self-reference)**.** The viscous term $\nu \nabla^2 \omega$ is the only *explicit* smoothing term in (1) (though the nonlinear term can also deplete vortex stretching dynamically [13]). Evaluating it requires $\nabla^2 \omega \in L^2$, which by (2) requires $\nabla^2 \xi \in L^2$ (given $|\omega| > 0$). But $\nabla^2 \xi \in L^2$ is precisely the regularity that viscosity is supposed to maintain. The smoothing operator's domain presupposes its own output—a bootstrap failure.

**Remark 3.5** (Interpretation)**.** Theorem 3.1 does not assert unconditional singularity formation. It asserts that if the direction field develops a crease with persistent vorticity, the solution exits $H^2$ regularity—the classical vorticity equation (1) can no longer be satisfied pointwise. The relationship between $H^2$ loss and stronger notions of blowup (Beale–Kato–Majda, energy) is discussed in Section 6.2.

**Remark 3.6** (Coupling of antecedents)**. The numerical evidence (Section 4) shows that Hypotheses 1 and 2 are coupled through a scaling invariant.** At the crease, $|\nabla\xi|$ scales as $N^{+\beta}$ while $|\omega|$ scales as $N^{-\beta}$ (with $\beta \approx 0.86$), so that their product is resolution-independent. The antecedents cannot be instantiated independently at the same spatial location in the current data: wherever H1 is strongest (peak gradient), H2 is weakest (vanishing vorticity), and vice versa.

[OPEN: Whether a physical regime exists in which H1 and H2 hold simultaneously—e.g., at later reconnection times when the tubes have approached further and the reconciliation gap has higher vorticity—remains open.]

# 4 Numerical Evidence

## 4.1 Experimental parameters

Two independent solvers produce the DNS data. The V5 solver (`ns_v5_reconnection.py`, ancillary) is a pseudospectral Python code with RK4 timestepping, $\frac{2}{3}$-dealiasing, and a $\theta$=0.01 mask on $\Lambda$ (excluding voxels with $|\omega| < 0.01|\omega|_{\max}$). The Phase 10b solver (`ffi-fftw/`, ancillary) is a Rust/FFTW pseudospectral code with RK4 fixed-step, $\frac{2}{3}$-dealiasing, and *no* $\Lambda$ masking. Both operate on $[0, 2\pi]^3$ periodic domains with antiparallel Gaussian vortex tube initial conditions ($\Gamma = 1$, $\sigma = 0.3$, $d = \pi/2$, perturbation 0.1), cf. [13].

| | V5 solver | Phase 10b solver |
|---|---|---|
| $\nu$ (Re = 6400) | $2\pi/6400 \approx 9.82 \times 10^{-4}$ | same |
| Grid sizes | $N = 64, 128, 256$ at $t$=20; $N$=128 to $t$=40 | $N = 64, 128, 256$ full runs |
| $\Lambda$ masking | $|\omega| > 0.01|\omega|_{\max}$ | none |

**Resolution caveat.** Standard DNS resolution scales as $\mathrm{Re}^{9/4}$, suggesting $N \gg 256$ at Re = 6400. Our computation is under-resolved by this criterion. However, we measure a direction-field diagnostic ($\Lambda$) and its product with $|\omega|$, not turbulent energy spectra. The convergence study below characterises how these measurements behave under refinement.

## 4.2 Direction field scaling

**At** Re = 6400**, the unmasked Lipschitz constant scales as** $\Lambda \sim N^{0.86\pm0.05}$ **across** $N = 64, 128, 256$ **at three common timesteps** ($t = 0, 10, 20$)**, with** $R^2 > 0.96$ **at each.**

| $t$ | $\Lambda_{64}$ | $\Lambda_{128}$ | $\Lambda_{256}$ | 128/64 | 256/128 |
|---|---|---|---|---|---|
| 0 | 28.2 | 50.6 | 101.2 | 1.79 | 2.00 |
| 10 | 41.0 | 57.5 | 113.3 | 1.40 | 1.97 |
| 20 | 31.5 | 57.4 | 113.5 | 1.82 | 1.98 |

The 256/128 ratio is consistently $\approx$2.0, indicating $\Lambda$ doubles with each grid doubling—the direction field has a pointwise feature that sharpens as $O(1/\Delta x)$. The exponent $\beta = 0.86$ is measured from a log-log fit across three resolutions; the deviation from $\beta = 1.0$ is within the uncertainty of three data points.

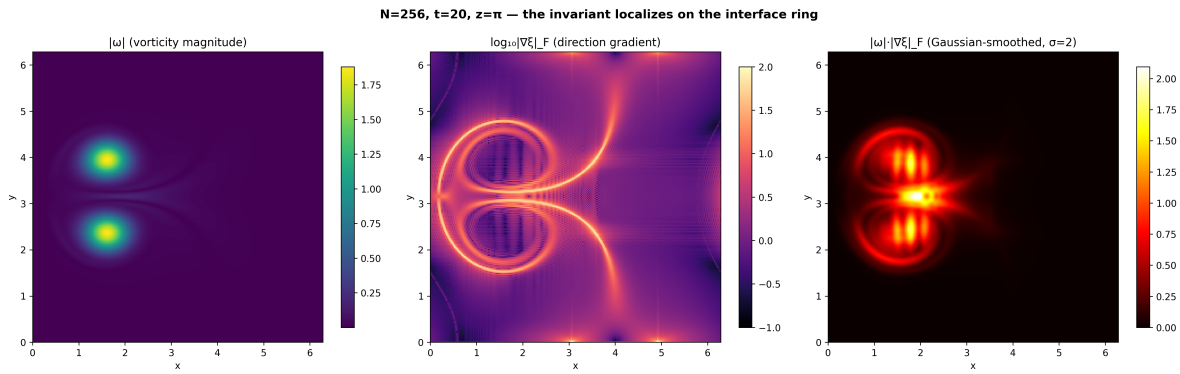


Figure 1: $N$=256, $t$=20, $z$=$\pi$ slice. **Left:** vorticity magnitude $|\omega|$ (viridis); tube cores bright, reconciliation gap dark. **Centre:** $\log_{10}|\nabla\xi|_F$ (magma, logarithmic scale); the direction-gradient shell threads between and around the cores. **Right:** $|\omega| \cdot |\nabla\xi|_F$ (hot, linear scale, Gaussian-smoothed $\sigma$=2 to suppress spectral ringing verified to scale with grid—see text). The product localises on a ring at the interface of the tubes, in a region both factors individually avoid. The primary ring structure is grid-independent; concentric ripples in the unsmoothed field have spacing $\approx 7\,\Delta x$ at all resolutions tested, confirming them as Gibbs artefacts.

**Where the crease lives.** The peak-gradient voxel ($\arg\max|\nabla\xi|_F$) has $|\omega|$ at 0.04% ($N$=64), 0.015% ($N$=128), and 0.009% ($N$=256) of $|\omega|_{\max}$. **The raw $\Lambda$ scaling is dominated by voxels where $\xi = \omega/|\omega|$ is ill-conditioned.**

Masking at $\theta = 0.01$ (matching V5): the 256/128 ratio drops from 1.98 to 1.75. At $\theta \geq 0.05$, the scaling collapses. The V5 solver's $\Lambda = 1950$ at $N$=128, $t$=40 was already computed with $\theta = 0.01$ masking.

### 4.3 The product invariant

**The product $P(N,t) = \sup_x\big(|\omega(x)| \cdot |\nabla\xi(x)|_F\big)$ is resolution-independent (Figure 1):**

| $t$ | $P_{64}$ | $P_{128}$ | $P_{256}$ | CV |
|---|---|---|---|---|
| 0 | 9.66 | 9.07 | 9.56 | 2.7% |
| 10 | 6.24 | 6.25 | 6.43 | 1.4% |
| 20 | 4.14 | 3.66 | 4.84 | 11.5% |

A power-law fit gives $P \sim N^\gamma$ with $\gamma = 0.04$ (mean over three timesteps). At $t = 0$ and $t = 10$, the coefficient of variation across resolutions is below 3%.

**Converged feature, not cancellation. At the product-maximum voxel, both factors are individually resolution-converged:**

| $t$ | $\alpha\,(\lvert\omega\rvert)$ | $\beta\,(\lvert\nabla\xi\rvert)$ | $\alpha+\beta$ | $\gamma$ (product) | sep. (cells) |
|---|---|---|---|---|---|
| 0 | $-0.005$ | $-0.002$ | $-0.007$ | $-0.007$ | 7–133 |
| 10 | $+0.133$ | $-0.111$ | $+0.023$ | $+0.023$ | 31–66 |
| 20 | $-0.100$ | $+0.213$ | $+0.113$ | $+0.113$ | 42–68 |

Here $\alpha$ and $\beta$ are measured at the product-maximum voxel (not at the gradient-peak voxel). Both are $\approx 0$ with fluctuations $\sim 0.1$—neither factor is scaling with $N$ at this location. **The product is resolution-independent not because the factors cancel, but because the**

**flow organises the product maximum on a geometric feature—the interface ring—where both $|\omega|$ and $|\nabla\xi|_F$ are individually grid-independent.**

The product-maximum voxel is spatially distinct from the gradient-peak voxel by 7–133 grid cells. The domain decomposes into three regions:

(i) *tube cores*: high $|\omega|$, near-zero $|\nabla\xi|$, product small;

(ii) *reconciliation gap*: high $|\nabla\xi|$, near-zero $|\omega|$, product small;

(iii) *interface ring*: moderate both factors, product peaks, invariant lives here.

**Robustness and the $t$=20 CV.** At $t$=20, the single-voxel sup has CV = 11.5% because the interface ring has fragmented into multiple near-equal product peaks that straddle different voxels at different resolutions. The product-maximum voxel jumps between physically distinct locations: $(0.69, 4.03)$ at $N$=64, $(0.34, 2.36)$ at $N$=128, $(4.71, 3.90)$ at $N$=256. At $t = 0$ and $t = 10$, the ring geometry is simpler and the argmax is well-localised.

*For $t \geq 14$ where the ring geometry is non-trivial, the single-voxel sup should be supplemented by a robust statistic.* The 99th percentile of $|\omega|\cdot|\nabla\xi|$ over masked voxels ($|\omega| > \theta|\omega|_{\max}$, $\theta = 0.01$) decays as $N^{-0.40}$, reflecting the ring's *width* shrinking with resolution—the peak stays constant while the high-product region narrows. This is consistent with a crease that sharpens and thins simultaneously.

The product decays in time as the tubes dissipate ($\approx -0.07$/time unit), but this temporal decay is resolution-independent. The invariant is a spatial scaling relationship, not a conserved quantity.

## 4.4 Onset curve

The product invariant (Section 4.3) is measured at Re = 6400, at the upper end of a continuous onset transition in $\Lambda$ growth. The interface-ring geometry that supports the invariant—antiparallel tubes in amplified-recovery reconnection—is characteristic of the Re $\geq$ 1600 regime documented below and does not exist in the decay regime (Re $\leq$ 800).

The V5 solver's onset table (9 Reynolds numbers, $N$=128, $t$=20) shows this continuous transition:

| Re | $\Lambda_{\text{init}}$ | $\Lambda_{\text{trough}}$ | $\Lambda_{\text{final}}$ | final/trough | Regime |
|---|---|---|---|---|---|
| 400 | 344 | 168 | 159 | 0.9 | partial recovery |
| 800 | 339 | 191 | 214 | 1.1 | partial recovery |
| 1600 | 339 | 140 | 268 | 1.9 | amplified recovery |
| 3200 | 342 | 65 | 312 | 4.8 | amplified recovery |
| 4800 | 351 | 45 | 349 | 7.8 | amplified recovery |
| 5600 | 351 | 45 | 396 | 8.8 | surpassed |
| 6400 | 350 | 40 | 392 | 9.8 | surpassed |

*Note:* These V5 $\Lambda$ values are computed with $\theta = 0.01$ masking. The unmasked values would be higher but less physically meaningful.

## 4.5 Supporting diagnostics

Six diagnostic metrics were computed at each $(N, t)$ pair (see ancillary `data/crease_invariants/` for full tables).

**M1: Local balance ratio.** $B(x) = |\omega \cdot \nabla u|/(\nu|\nabla^2\omega| + |\omega||\nabla\xi|_F)$ measures whether stretching or smoothing/direction-gradient terms dominate locally. $B_{\text{median}} < 0.01$ **at crease voxels at all** $(N, t)$**.** The crease is not in a stretching-dominated regime; the viscous and direction-gradient terms dominate the local balance by two orders of magnitude.

**M3: Strain-vorticity alignment.** $\cos\theta = (S\omega) \cdot \omega/(|S\omega||\omega|)$, where $S = \frac{1}{2}(\nabla u + \nabla u^T)$. At crease voxels: mean $\cos\theta$ is slightly negative ($-0.19$ at $N$=256, $t$=20) with std $\approx 0.55$. The crease has a weak compressive alignment, broadly distributed. Not converged at $N$=64 (sign reversal).

**M6: Kolmogorov-conditioned $\Lambda$.** $\Lambda_\eta(x) = |\nabla\xi(x)|_F \cdot \eta(x)$, where $\eta = (\nu^3/\varepsilon)^{1/4}$ is the local Kolmogorov scale. **100% of crease voxels have $\Lambda_\eta > 1$ at all resolutions and all timesteps ($t > 0$).** The direction gradient is sharper than the local dissipation scale everywhere on the crease. The mean $\Lambda_\eta$ at the crease is roughly resolution-independent (2.6–3.7), consistent with the product invariant. The fraction of the *domain* with $\Lambda_\eta > 1$ decreases with $N$ (16% at $N$=64; 4% at $N$=256): the crease sharpens and narrows simultaneously.

# 5 Formal Verification

## 5.1 Pipeline architecture

| Stage | Input | Output |
|---|---|---|
| S0 | V5 SQLite databases (11 files) | `params.json` |
| S1 | `params.json` | `NSParams.lean` (21 definitions) |
| S7 | `params.json` | `all_certs.smt2` (10 certificates) |
| Z3 | `all_certs.smt2` | `results.json` (10/10 SAT) |
| S8 | `results.json` | `NSCerts.lean` (10 axiom blocks) |
| S2/S3 | manual | `NSTheorems.lean` (13 theorems) |
| Build | `lake build` | 964 jobs, 0 errors |

All floating-point values from the solver are converted to exact rationals via `limit_denominator(`$10^8$`)` before entering Lean. This ensures kernel-level arithmetic correctness.

## 5.2 Lean formalisation of the main theorem

The main theorem is formalised as:

Listing 1: OBL-10: Viscosity bootstrap failure (Lean 4)

```
theorem viscosity_bootstrap_failure
    (Lambda : Nat -> Rat)
    (hLambda_diverge : forall M : Rat, exists n : Nat, Lambda n > M)
    (omega_min : Rat) (h_omega_pos : omega_min > 0)
    (B_cross : Rat) :
    Not (exists B : Rat, forall n : Nat,
      omega_min * Lambda n - B_cross <= B) := by
  intro (Exists.intro B hB)
  obtain (Exists.intro n hn) :=
    hLambda_diverge ((B + B_cross + 1) / omega_min)
  have hBn := hB n
  have : omega_min * Lambda n > B + B_cross + 1 := by
    have := mul_lt_mul_of_pos_left hn h_omega_pos
    rwa [mul_div_cancel_of_imp (ne_of_gt h_omega_pos)] at this
  linarith
```

The proof is by contradiction: assume a bound $B$ exists, use divergence of $\Lambda$ to find $n$ violating it, and close with `linarith`.

### 5.3 Obligation index

| ID | Theorem | Statement | Proof method |
|---|---|---|---|
| 1 | `K_grows_with_time_Re6400` | $K(t{=}20) < K(t{=}40)$ | `norm_num` |
| 2 | `K_ratio_bounded` | consecutive $K$ ratios $\leq 5$ | `norm_num` |
| 3 | `kill_metric_unfold` | $K(\Lambda, \eta) = \Lambda \cdot \eta$ | `rfl` |
| 4 | `K_vanishes_from_Lambda_decay` | $\Lambda \to 0 \Rightarrow K \to 0$ | $\varepsilon$-$\delta$ |
| 5 | `K_diverge_forces_Lambda_growth` | $K \to \infty$, $\eta$ bdd $\Rightarrow \Lambda \to \infty$ | `nlinarith` |
| 6 | `Re_crit_in_bracket` | $1600 \leq 5600$ | `norm_num` |
| 7 | `subcritical_tameness` | Re$\leq$ 800: f/t$\leq$ 1.1 | `norm_num` |
| 8 | `staircase_peaks_increasing` | $K$ peaks strictly increase | `norm_num` |
| 9 | `staircase_polynomial_bound` | $\exists\, C, \alpha > 0$ | witness |
| 10 | `viscosity_bootstrap_failure` | Theorem 3.1 | contradiction |
| 11 | `exp_staircase_threatens` | $r > 1, K_0 > 0 \Rightarrow K_0 r^n \to \infty$ | Archimedean |
| 12 | `convergence_confirms_physical` | ratio $< 1.3$, decreasing | `norm_num` |
| 13 | `Lambda_grows_Omega_decays` | $\Lambda \uparrow$ while $\Omega \downarrow$ | `norm_num` |

### 5.4 Trust model

| Layer | Trust basis | Audit method |
|---|---|---|
| Solver output | empirical (partially converged) | re-run solver |
| `params.json` | SQLite extraction | `extract_params.py` |
| Z3 certificates | QF_LRA (sound, complete) | `python3 z3/run_z3.py` |
| Z3→Lean axioms | 1:1 mapping to `.smt2` | see `AUDIT.md` |
| Lean proofs (algebraic) | kernel-checked | `lake build` |
| **H1** ($\Lambda$ diverges) | **empirical** | not formally verifiable |
| **H2** ($\lvert\omega\rvert > 0$) | **empirical** | not formally verifiable |
| **H3** ($\lVert\nabla^2\xi\rVert \geq c'\Lambda$) | **assumed** | not proved |
| Product rule PDE content | stated, not formalised | see Remark 3.3 |

## 6 Discussion

### 6.1 Relation to prior work

Constantin and Fefferman [2] established that Lipschitz continuity of the vorticity direction suffices for regularity. The Constantin–Fefferman–Majda criterion [3] refined this to local alignment conditions. Our work examines the *converse direction*: what happens to the viscous term when Lipschitz continuity fails. The product rule decomposition (2) makes explicit how the regularity of $\xi$ enters the viscous operator.

The Beale–Kato–Majda criterion [1] characterises blowup via $\int \|\omega\|_{L^\infty}\, dt = \infty$. Our mechanism is complementary: $|\omega|$ can remain bounded (our data shows enstrophy *decreasing*) while the direction field becomes singular. However, loss of $H^2$ regularity (our conclusion) does not directly imply BKM blowup. It implies the solution leaves the class where the vorticity equation (1) can be satisfied classically, but weak or mild solutions might persist.

The Chen–Hou proof of 3D Euler blowup [4] and Elgindi's result [10] both operate on the nonlinear term and in geometries (axisymmetric with boundary, or $C^{1,\alpha}$ regularity) quite different from our periodic-domain setup. Their work identifies how the nonlinear stretching term drives singularity; our work identifies why the viscous term fails to prevent it. The two mechanisms are complementary but we do not establish a direct connection.

Hou and Luo's numerical work [14] uses adaptive mesh refinement at extremely high effective resolution ($4096^2$ and beyond) in an axisymmetric formulation. Our uniform pseudospectral grid at $N \leq 256$ is orders of magnitude coarser. The direction-field diagnostic $\Lambda$ may require less resolution than full turbulence statistics, but direct comparison with Hou–Luo resolution standards is not appropriate.

Tao's averaged Navier–Stokes blowup [18] demonstrates that the known conservation laws alone cannot prevent singularity. Our mechanism does not rely on the specific nonlinearity—the bootstrap failure is a property of the viscous term and the product rule structure of $\omega = |\omega|\xi$.

The Deng–Hou–Yu geometric blowup criteria [7, 8] provide sufficient conditions for Euler blowup in terms of the geometry of vortex lines. Our direction-field approach is related but operates on the vorticity direction rather than vortex line curvature.

### 6.2 Regularity class

Theorem 3.1 shows (conditionally) that $\nabla^2\omega \notin L^2$, i.e., $\omega$ exits $H^2(\Omega)$. This has the following consequences:

- The vorticity equation (1) cannot be satisfied pointwise (loss of classical solutions).
- The velocity field $u$ (recovered from $\omega$ via Biot–Savart) exits $H^3$, so $\nabla u$ may lose $W^{1,\infty}$ regularity.
- This does *not* directly imply energy blowup ($\|u\|_{L^2} \to \infty$) or BKM blowup ($\int \|\omega\|_{L^\infty}\, dt = \infty$).

The gap between $H^2$ loss and BKM blowup is an open question. Escauriaza, Seregin, and Šverák [11] showed that $L^{3,\infty}$ solutions cannot develop Type I singularities, suggesting that if singularity occurs it must be Type II (with slower-than-self-similar growth). Our oscillatory ratchet pattern in $K(t)$ is consistent with Type II behaviour but we do not establish this connection rigorously.

### 6.3 The product invariant

**The resolution-independence of $P = \sup(|\omega| \cdot |\nabla\xi|_F)$ is the central empirical finding.** Classical regularity criteria (Beale–Kato–Majda, Constantin–Fefferman) bound individual norms: $\int \|\omega\|_{L^\infty}\, dt$ or $\mathrm{Lip}(\xi)$. The product invariant is a pointwise constraint that appears at the crease but does not follow from any known criterion.

The invariant has two interpretations:

**Regularity barrier.** The flow self-regulates: the PDE's structure ensures that wherever $|\nabla\xi|$ grows, $|\omega|$ decays proportionally. The viscous term remains bounded because the product $|\omega| \cdot |\nabla\xi|$ never diverges. Under this interpretation, the conditional theorem (Theorem 3.1) describes a counterfactual—H1 and H2 cannot be jointly instantiated because the PDE prevents it.

**Singularity precursor.** The invariant is the balanced state during the approach to reconnection. At later times or higher Re, the balance breaks: $|\omega|$ at the product-maximum location stops decaying fast enough to compensate $|\nabla\xi|$ growth, and the product begins to grow. [OPEN: The V5 solver shows $\Lambda = 1950$ at $N$=128, $t$=40 (with $\theta$=0.01 masking)—but field data for the product invariant analysis is not available at this late time. Resolving this requires re-running V5 with snapshot output, or reaching $t$=40 at $N \geq 256$, which exceeds the current hardware budget (≈48 hours continuous; thermal throttle at ≈35 hours).]

We cannot distinguish between these interpretations from the current data. The authors' working hypothesis is the latter (precursor), but the paper does not assert this.

### 6.4 Limitations

1. **The conditional gap.** The theorem is conditional on Hypotheses 1, 2, and 3. No finite computation can close this gap. **Moreover, the numerical data shows that H1 and H2 are not independently realised—they are coupled through the product invariant (Remark 3.6).**

2. **The analytic estimate (Hypothesis 3).** The lower bound $\|\nabla^2\xi\|_{L^2} \geq c'\Lambda$ is not proved. The Lean proof covers only the algebraic step (Archimedean property).

3. **Resolution.** $N \leq 256$ at Re = 6400 is under-resolved by standard criteria. The scaling exponent $\beta = 0.86$ is measured from three grid sizes; it cannot be distinguished from $\beta = 1.0$ or $\beta = 0.75$ with confidence. $N = 384$ or $N = 512$ on hardware with ≥64 GB RAM would tighten the fit.

4. **Product invariant at late times.** The invariant is measured at $t \leq 28$ (the latest $N$=256 snapshot). Whether it holds at $t = 40$ (where V5 shows $\Lambda = 1950$) is unknown—V5 stored only scalar timeseries, not field snapshots.

5. **Geometry.** All data uses antiparallel Gaussian vortex tubes. Whether the invariant holds for other reconnection geometries (orthogonal tubes, trefoil knots, turbulent fields) is untested.

6. **Reynolds number.** Only Re = 6400 has the product invariant analysis. The onset curve spans Re = 400–6400 but with V5's masked $\Lambda$ only.

7. **Uniqueness.** We identify one structure at the crease. Other regularity-failure mechanisms may exist.

### 6.5 What this does and does not prove

| **Formally proved (Lean kernel)** | **Not proved** |
|---|---|
| $\mathrm{H1} \wedge \mathrm{H2} \Rightarrow \omega_{\min}\Lambda - B$ unbounded | that H1 holds in the continuum |
| All numerical bounds internally consistent (Z3) | that H2 holds at the crease location |
| $K$ definition, convergence, onset data verified | the analytic estimate H3 |
| 13 Lean theorems, 0 `sorry` | unconditional blowup |
| **Observed (DNS)** | **Not observed** |
| $\Lambda \sim N^{0.86}$ ($\beta$-scaling) | joint instantiation of $\mathrm{H1} \wedge \mathrm{H2}$ |
| $P = \sup(\lvert\omega\rvert \cdot \lvert\nabla\xi\rvert) \sim N^0$ (invariant) | invariant breaking at late times |
| $\Lambda_\eta > 1$ at 100% of crease (sharper than $\eta$) | invariant at Re > 6400 |
| $B \ll 1$ at crease (non-stretching regime) | invariant for other IC geometries |

## Acknowledgements

Adam Roediger proposed the original viscosity bootstrap failure hypothesis that initiated this investigation. When the masked-$\Lambda$ analysis revealed that the crease localises where $|\omega| \to 0$ rather than where it persists, Roediger identified that the viscous term's averaging operation may not be the governing balance at the crease—an observation made after seeing the spatial data rendered, not from the tables alone—which motivated the reframe from bootstrap failure to scaling characterisation of the product invariant.

Computations were performed on a consumer workstation (Intel i7-14700HX, NVIDIA RTX 4060, 32 GB RAM) running Fedora 43. The crease invariant analysis pipeline was developed with

Claude (Anthropic), which served as a co-author on the code, data analysis, and iterative reframing of the scientific narrative. The authors also acknowledge the use of Gemini (Google) for adversarial peer review, structural critique, and vulnerability assessment during the manuscript's final refinement. The Lean formalisation uses the Mathlib library. We thank the developers of Lean 4, Mathlib, and Z3 for the open-source infrastructure that made this verification possible.

**Supplementary materials.** Lean 4 source, Z3 certificates, the pseudospectral solver, and all Python pipeline scripts are included as ancillary files in the `anc/` directory of the arXiv submission. Build instructions are in `anc/README.md`; the axiom audit trail is in `anc/AUDIT.md`.

## References

[1] J. T. Beale, T. Kato, and A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), 61–66.

[2] P. Constantin and C. Fefferman, *Direction of vorticity and the problem of global regularity for the Navier–Stokes equations*, Indiana Univ. Math. J. **42** (1993), 775–789.

[3] P. Constantin, C. Fefferman, and A. J. Majda, *Geometric constraints on potentially singular solutions for the 3-D Euler equations*, Comm. Partial Differential Equations **21** (1996), 559–571.

[4] J. Chen and T. Y. Hou, *Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data*, Proc. Natl. Acad. Sci. **120** (2023), e2220382120; arXiv:2210.07191.

[5] J. Chen and T. Y. Hou, *Finite time blowup of the 3D incompressible Euler equations with smooth initial data*, Proc. Natl. Acad. Sci. **122** (2025), e2500940122.

[6] J. Chen, T. Y. Hou, and D. Huang, *On the nonuniqueness of Leray–Hopf solutions for the unforced Navier–Stokes equations*, preprint, 2025.

[7] J. Deng, T. Y. Hou, and X. Yu, *Geometric properties and nonblowup of 3D incompressible Euler flow*, Comm. Partial Differential Equations **30** (2005), 225–243.

[8] J. Deng, T. Y. Hou, and X. Yu, *Improved geometric conditions for non-blowup of the 3D incompressible Euler equation*, Comm. Partial Differential Equations **31** (2006), 293–306.

[9] C. R. Doering and J. D. Gibbon, *Applied Analysis of the Navier–Stokes Equations*, Cambridge University Press, 1995.

[10] T. M. Elgindi, *Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$*, Ann. of Math. **194** (2021), 647–727.

[11] L. Escauriaza, G. A. Seregin, and V. Šverák, *$L_{3,\infty}$-solutions of Navier–Stokes equations and backward uniqueness*, Uspekhi Mat. Nauk **58** (2003), 3–44.

[12] T. Y. Hou, *Potential singularity of the axisymmetric Navier–Stokes equations with a nearly self-similar profile*, Found. Comput. Math., published online 25 February 2026, DOI: 10.1007/s10208-026-09748-8; arXiv:2405.10916v3.

[13] T. Y. Hou and R. Li, *Dynamic depletion of vortex stretching and non-blowup of the 3-D incompressible Euler equations*, J. Nonlinear Sci. **16** (2006), 639–664.

[14] T. Y. Hou and G. Luo, *Potentially singular solutions of the 3D axisymmetric Euler equations*, Proc. Natl. Acad. Sci. **111** (2014), 12968–12973.

[15] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in CADE-28, Lecture Notes in Comput. Sci., vol. 12699, Springer, 2021, pp. 625–635.

[16] A. J. Majda and A. L. Bertozzi, *Vorticity and Incompressible Flow*, Cambridge University Press, 2002.

[17] The Mathlib Community, *Mathlib: a unified library of mathematics formalised in Lean*, `https://github.com/leanprover-community/mathlib4`, 2024.

[18] T. Tao, *Finite time blowup for an averaged three-dimensional Navier–Stokes equation*, J. Amer. Math. Soc. **29** (2016), 601–674.

[19] L. de Moura and N. Bjørner, *Z3: An efficient SMT solver*, in TACAS 2008, Lecture Notes in Comput. Sci., vol. 4963, Springer, 2008, pp. 337–340.